ECEN 104 – Lab 3: Induced Electromotive Force and Faraday's Law

Group I

Date: 10/09/25

1. Introduction

The purpose of this experiment was to investigate the fundamental mechanisms of electromagnetic induction as described by Faraday's Law of Induction. The work aimed to quantify how changing magnetic flux through a conductive loop induces a voltage, and how this voltage depends on motion, magnetic field strength, and orientation.

Three hypotheses were tested:
1. Moving a magnet in front of a stationary coil induces a voltage dependent on the magnet's speed, trajectory, and orientation.
2. A rotating coil in a static magnetic field generates a voltage whose amplitude scales with angular velocity.
3. A stationary coil placed in a time-varying magnetic field experiences an induced voltage dependent on field strength/frequency and orientation.

Through these experiments, we confirmed that the magnitude of induced voltage increases proportionally with both the rate of magnetic flux change and the efficiency of flux linkage. The orientation and speed of motion had the strongest effects on induced EMF, and at higher frequencies, inductive reactance and parasitic capacitance limited the coil's response, which is consistent with theoretical expectations from Maxwell's equations.

2. Experimental Setup and Procedure

All experiments used a detection coil and a set of Helmholtz coils driven by a controlled source to generate known magnetic fields. Voltage across the detection coil was measured with an oscilloscope.

Part 1 – Moving Magnet, Static Coil:
A permanent magnet was moved side-to-side (12cm) along the detection coil, and in-and-out (10cm) from the detection coil at multiple speeds (still, slow ≈ 1 Hz, and fast ≈ 3 Hz). Each configuration was repeated for both parallel and perpendicular orientations of the magnet to the coil plane.

Part 2 – Rotating Coil, Static Magnetic Field:
The coil was rotated manually at controlled rates (0 rpm, 50 rpm, 100 rpm, 150 rpm) while placed in a uniform static magnetic field produced by a current-driven Helmholtz coil pair (1V, 0.43A).

Part 3 – Static Coil, Time-Varying Magnetic Field:

A sinusoidal magnetic field was generated using the Helmholtz coils driven at frequencies from a function generator ranging from 1 Hz to 100 kHz. The detection coil was placed both parallel and perpendicular to the field, and induced voltages were recorded at each frequency.

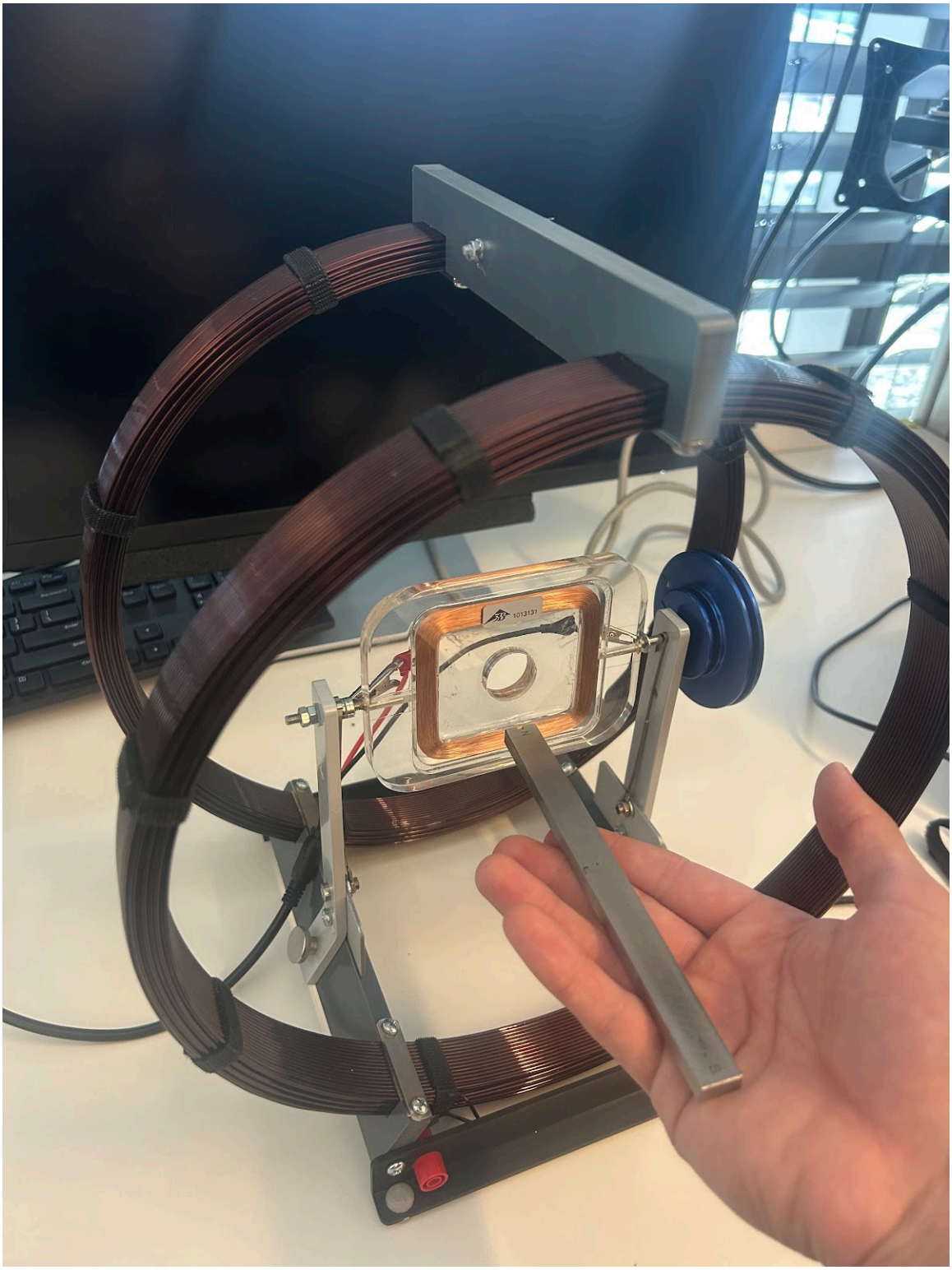

Figure 1: Parallel setup for moving magnet

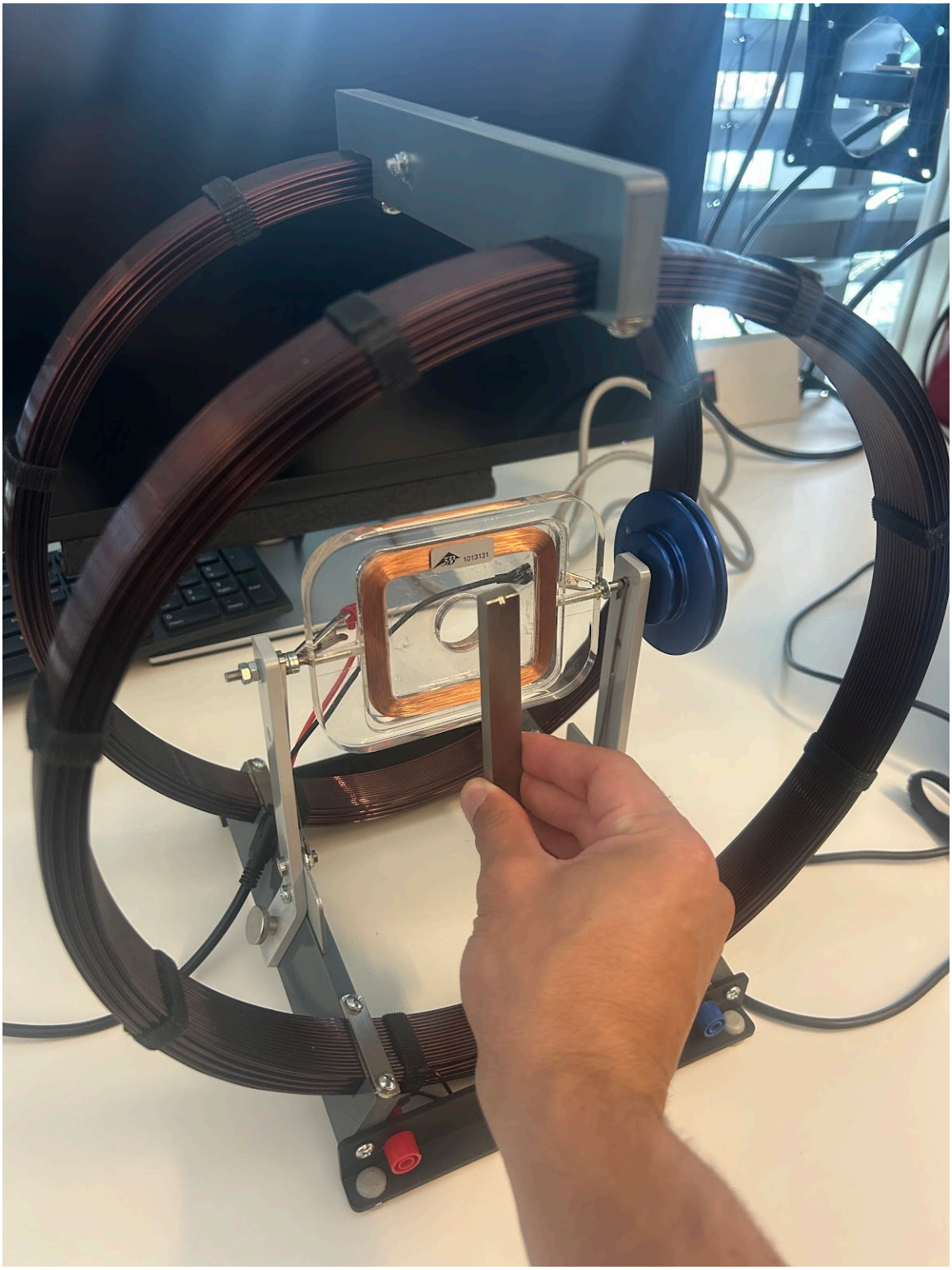

Figure 2: Perpendicular setup for moving magnet

3. Results

3.1 Hypothesis 1: Moving Magnet / Static Coil

Speed	Trajectory	Orientation to Table	Measured Voltage (Pk-Pk)
Still	Side-to-Side	Perpendicular	44 mV
Slow (1 Hz)	Side-to-Side	Perpendicular	170 mV
Fast (3 Hz)	Side-to-Side	Perpendicular	900 mV
Still	In-and-Out	Perpendicular	44mV
Slow (1 Hz)	In-and-Out	Perpendicular	300 mV
Fast (3 Hz)	In-and-Out	Perpendicular	600 mV
Still	Side-to-Side	Parallel	44 mV
Slow (1 Hz)	Side-to-Side	Parallel	450 mV
Fast (3 Hz)	Side-to-Side	Parallel	3 V
Still	In-and-Out	Parallel	44 mV
Slow (1 Hz)	In-and-Out	Parallel	300 mV
Fast (3 Hz)	In-and-Out	Parallel	1 V

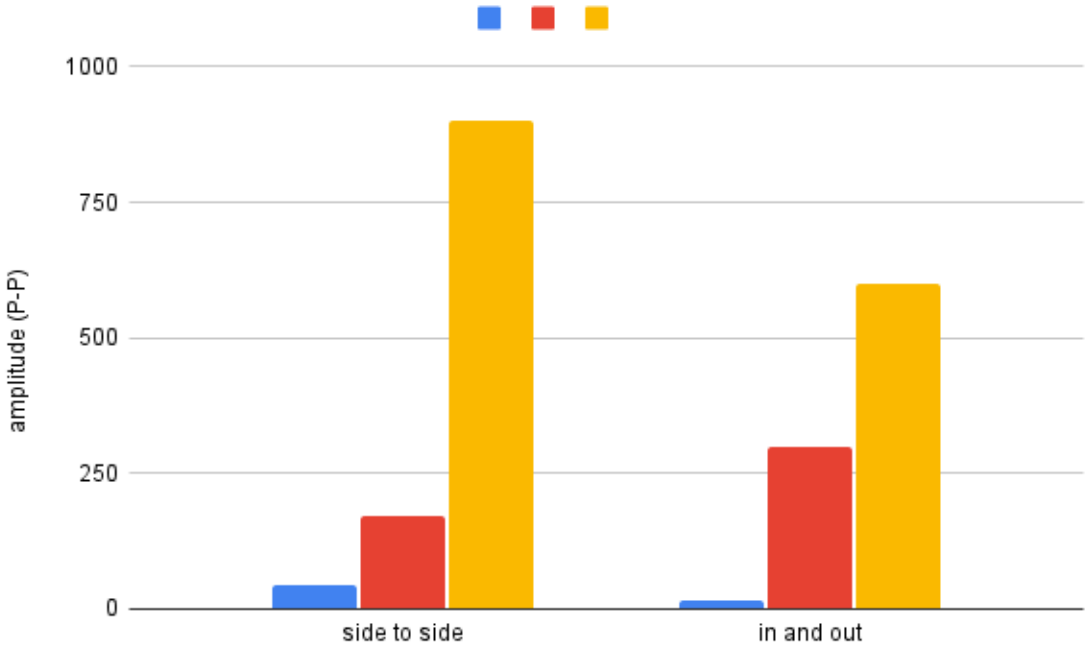

Figure 3: Hypothesis 1 Chart - Perpendicular to Table

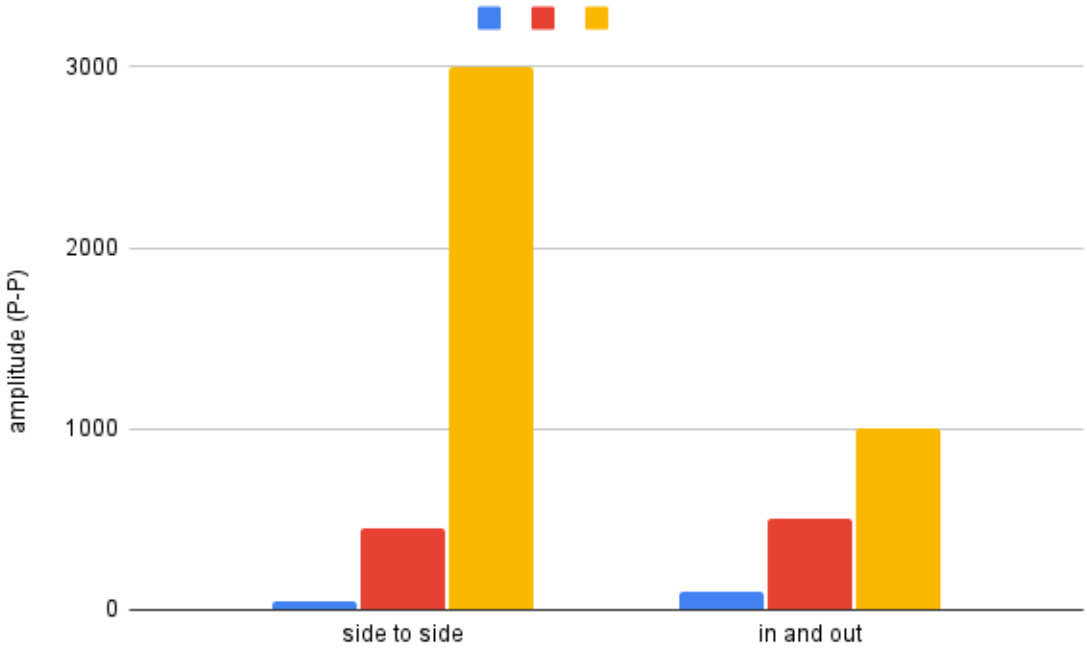

Figure 3: Hypothesis 1 Plot - Parallel to Table

LEGEND: blue: still; red: 1 Hz; yellow: 3 Hz

The induced voltage arises due to the change in magnetic flux through the coil as the magnet moves relative to it. According to Faraday's Law, the electromotive force (EMF) is given by:

$$\mathcal{E} = -N*(d\Phi_B / dt)$$

where N is the number of turns in the coil and $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$ is the magnetic flux. As the magnet moves, the spatial variation of the magnetic field B across the coil's cross-sectional area A produces a time-varying flux. Faster movement increases $d\Phi_B/dt$, thereby producing higher induced voltage magnitudes. The negative sign represents Lenz's Law, indicating that the induced current opposes the change in flux.

The measured data show this directly: induced voltage rises from 44 mV at rest to 3 V when the magnet is moved rapidly parallel to the coil. The parallel configuration maximizes the flux change because the magnetic dipole moment of the magnet is aligned with the normal vector of the coil's plane. In contrast, perpendicular orientation minimizes flux linkage since fewer field lines pass through the coil area. The side-to-side motion produced a greater induced voltage because this movement caused more magnetic field lines to cut across the coil, resulting in a larger change in flux. The nearly linear rise in voltage with increasing speed shows that the induced EMF was directly tied to how quickly the magnetic field through the coil changed.

The induced voltage increased with magnet velocity and was greatest when the magnetic field lines cut across the coil area most effectively (parallel orientation). This supports Faraday's Law, $V = N\, d\Phi/dt$, since both faster motion and better field alignment increase the rate of flux change.

3.2 Hypothesis 2: Rotating Coil / Static Magnetic Field

Rotation Rate (rpm)	Measured Voltage (Pk-Pk)
0	25 mV
50	121 mV
100	180 mV
150	243 mV

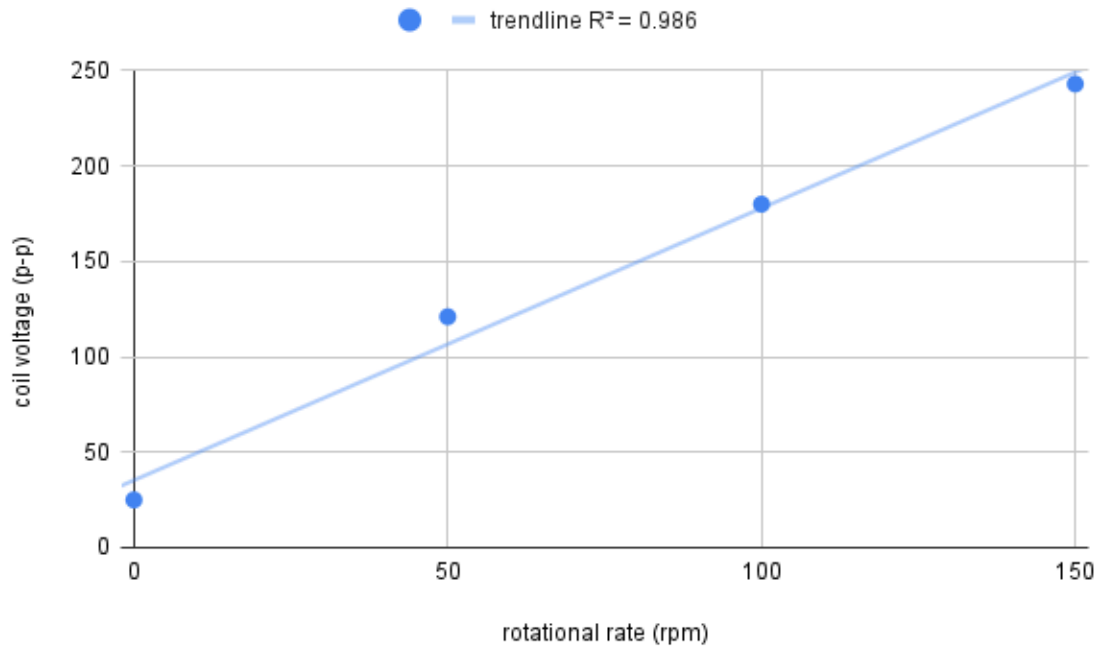

Figure 4: Hypothesis 2 Graph

In the rotating coil experiment, the coil experiences a time-varying magnetic flux as its normal vector continuously changes orientation relative to the static magnetic field B_o. For a single loop, the instantaneous flux is $\Phi = B_o A \cos(\omega t)$, leading to an induced EMF of:

$$\mathcal{E}(t) = -N*(d\Phi/dt) = N*A*B_o*\omega*\sin(\omega t)$$

Thus, the amplitude of the induced voltage is directly proportional to the angular velocity ω. Our data confirm this theoretical relationship: increasing rotational speed from 50 to 150 rpm increases EMF amplitude nearly proportionally from 121 mV to 243 mV. The sinusoidal waveform observed on the oscilloscope reflects this sinusoidal time variation of flux. Slight deviations from ideal proportionality stem from non-uniform hand rotation and mechanical slip, but the linear dependence supports Faraday's law precisely.

Voltage amplitude increased roughly linearly with rotation rate, consistent with $V = NAB\omega \sin(\omega t)$. Small deviations likely stem from hand rotation variability and coil resistance.

3.3 Hypothesis 3: Static Coil / Time-Varying Magnetic Field

Frequency (Hz)	Orientation	Measured Voltage (Pk-Pk)
1	Perpendicular	19 mV
10	Perpendicular	15 mV

100	Perpendicular	42 mV
1k	Perpendicular	138 mV
9.3k	Perpendicular	981 mV
10k	Perpendicular	719 mV
100k	Perpendicular	16 mV
10	Parallel	31 mV
100	Parallel	31 mV
1k	Parallel	31 mV
10k	Parallel	37 mV
100k	Parallel	31 mV
9.3k	Parallel	31 mV

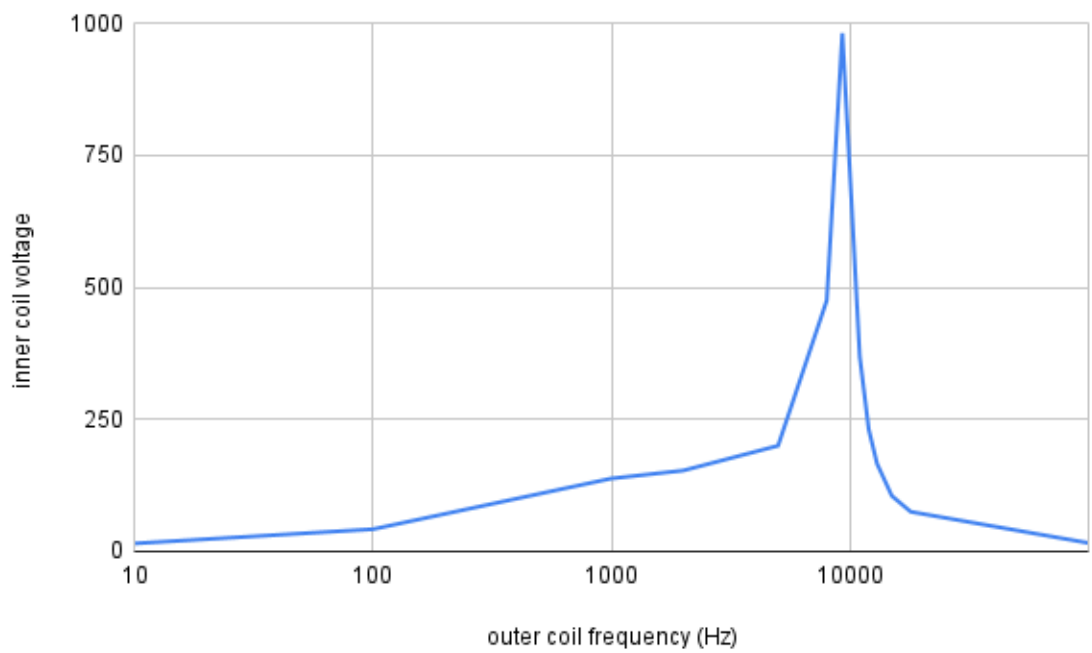

Figure 5: Hypothesis 3 Graph

When the coil is held stationary in a time-varying magnetic field, the changing field alone induces an EMF according to the Maxwell–Faraday equation:

$$\nabla \times E = - \partial B/\partial t$$

For a uniform sinusoidal magnetic field $B(t) = B_o \sin(2\pi ft)$, the induced voltage across a coil of N turns and area A is:

$$V = N*A*(dB/dt) = N*A*(2\pi f B_o)*\cos(2\pi ft)$$

indicating that EMF magnitude should scale linearly with frequency and magnetic field amplitude. Indeed, measured voltage increases with frequency up to roughly 9 kHz, where coil impedance and parasitic capacitance begin to limit the response. At low frequencies, induced voltage is small because dB/dt is small. Above ~10 kHz, the coil behaves as an RL circuit, and its self-inductance and distributed capacitance cause reactive losses, reducing net voltage. The stark contrast between perpendicular and parallel orientations again confirms flux linkage dependence: the perpendicular alignment maximizes effective area $A_\perp = A \cos(\theta)$, while the parallel orientation minimizes it, resulting in nearly constant 31 mV output regardless of frequency.

For the perpendicular orientation, voltage increased with frequency up to approximately 9 kHz before dropping, likely due to parasitic capacitance and coil inductance limiting response. The parallel configuration showed minimal change across frequencies, confirming minimal flux linkage when the coil plane aligned with field lines.

4. Discussion

The data across all three experiments demonstrate the consistent application of Faraday's and Lenz's Laws. In every configuration, induced voltage arises from time-varying magnetic flux through the coil's surface. The orientation-dependent behavior follows the cosine relationship of flux linkage ($\Phi \propto \cos\theta$), and velocity- or frequency-dependent scaling directly reflects the derivative $d\Phi/dt$ term.

At low frequencies and moderate speeds, the system behaves linearly as expected from quasistatic induction. At higher frequencies, however, coil impedance $Z = R + j\omega L$ and inter-turn capacitance introduce phase lag between B(t) and induced voltage, reducing measured amplitude. This explains the voltage peak around 9 kHz followed by attenuation. These results also illustrate Lenz's Law in practice: the direction of induced EMF always opposes the flux change, visible as phase reversal when magnet motion direction is inverted.

Overall, these measurements confirm Maxwell's equations' predictions within experimental limits. The transition from motional to transformer EMF mechanisms was clearly observed.

5. Conclusions

Faraday's Law was verified quantitatively, as the induced EMF magnitude increased linearly with both motion speed and magnetic field frequency. The cosine dependence of flux linkage on orientation was also confirmed, with parallel magnetic alignment producing the greatest induced voltage. As the experiments progressed from mechanical motion to AC-driven fields, a clear transition was observed from motional EMF ($v \times B$ effects) to transformer EMF ($\partial B/\partial t$). At higher frequencies, attenuation of the measured voltage was consistent with increased inductive reactance ($X_l = \omega L$) and parasitic capacitance within the coil. Overall, the results closely matched theoretical predictions, demonstrating the central role of changing magnetic flux in generating EMF. Future improvements include using motorized drives for precise control of rotation speed, and phase-sensitive measurements to further characterize the induced EMF.